

Lab 07 – Worksheet

Design Adder-Subtractor

Name:	ID:	Section:
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Task a) Selective 2's Complement

- (i) Complete the truth table of Selective 2's Complement in Table 7.1. As an example, few rows have been filled for you as hints.

Inputs to complement				Outputs of the complement		
O	B ₂	B ₁	B ₀	C ₂	C ₁	C ₀
0	0	0	0	0	0	0
0	0	0	1	0	0	1
0	0	1	0	0	1	0
0	0	1	1	0	1	1
0	1	0	0	1	0	0
1	0	0	0	0	0	0
1	0	0	1	1	1	1

Table 7.1: Truth table of the Selective 2's Complement module

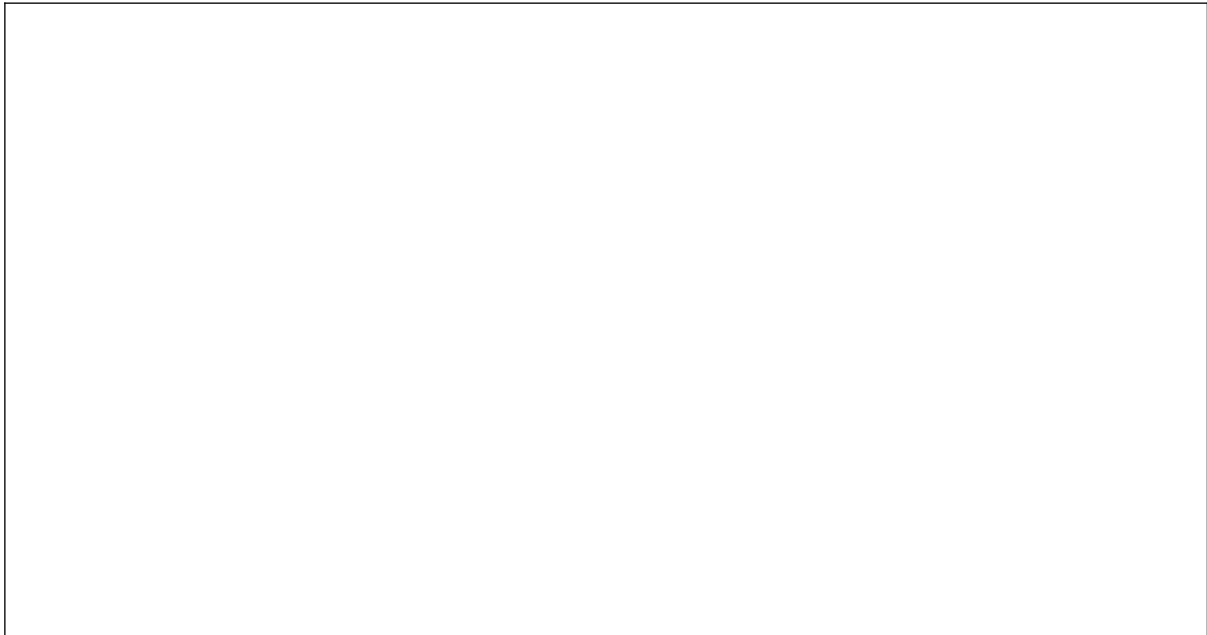
(ii) Write Boolean expression for each output in Table 7.2.

$C_2 =$	
$C_1 =$	
$C_0 =$	

Table 7. 2: Boolean expression for each output of Selective 2's Complement module

(iii) Provide appropriately commented code for your selective 2's complement design module.

(iv) Write a testbench to thoroughly test designed complement module.



- (v) Attach screenshot of Simulation output- make sure to scale properly for visibility of all case.



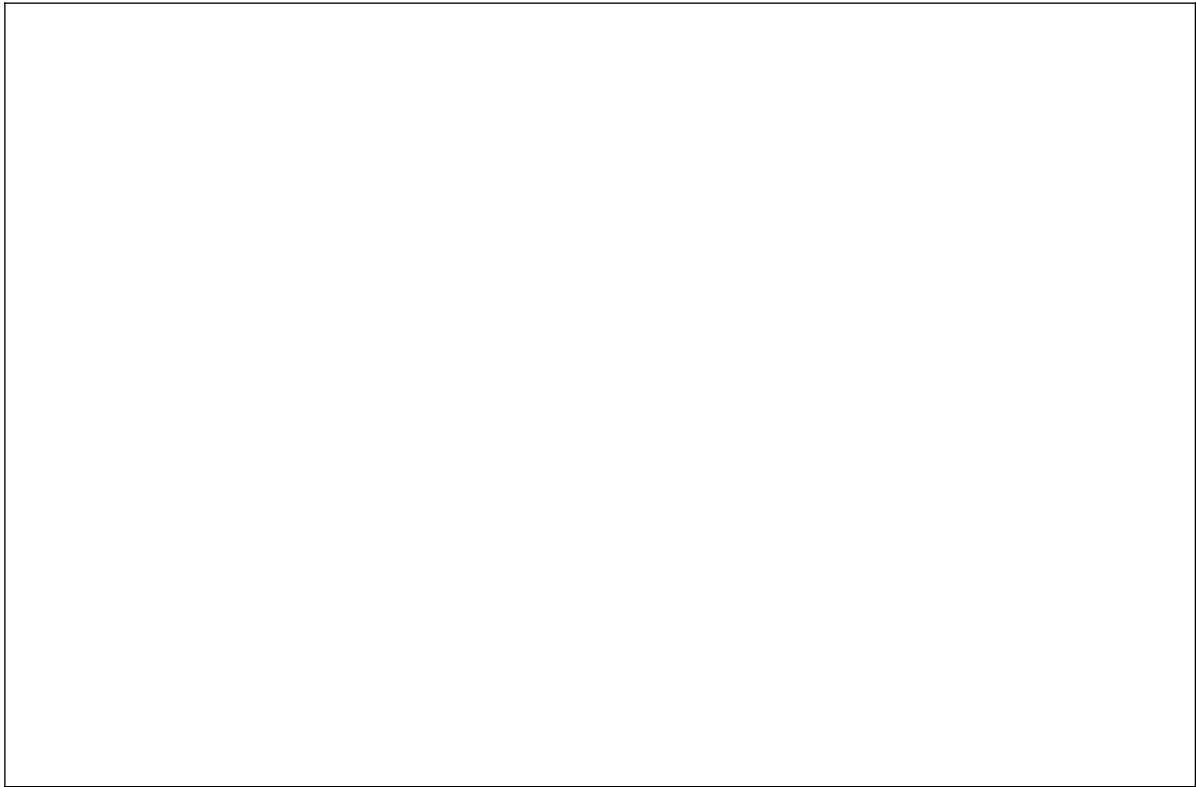
Task b) Adder/Subtractor System Integration

(i) Provide code for design module here.

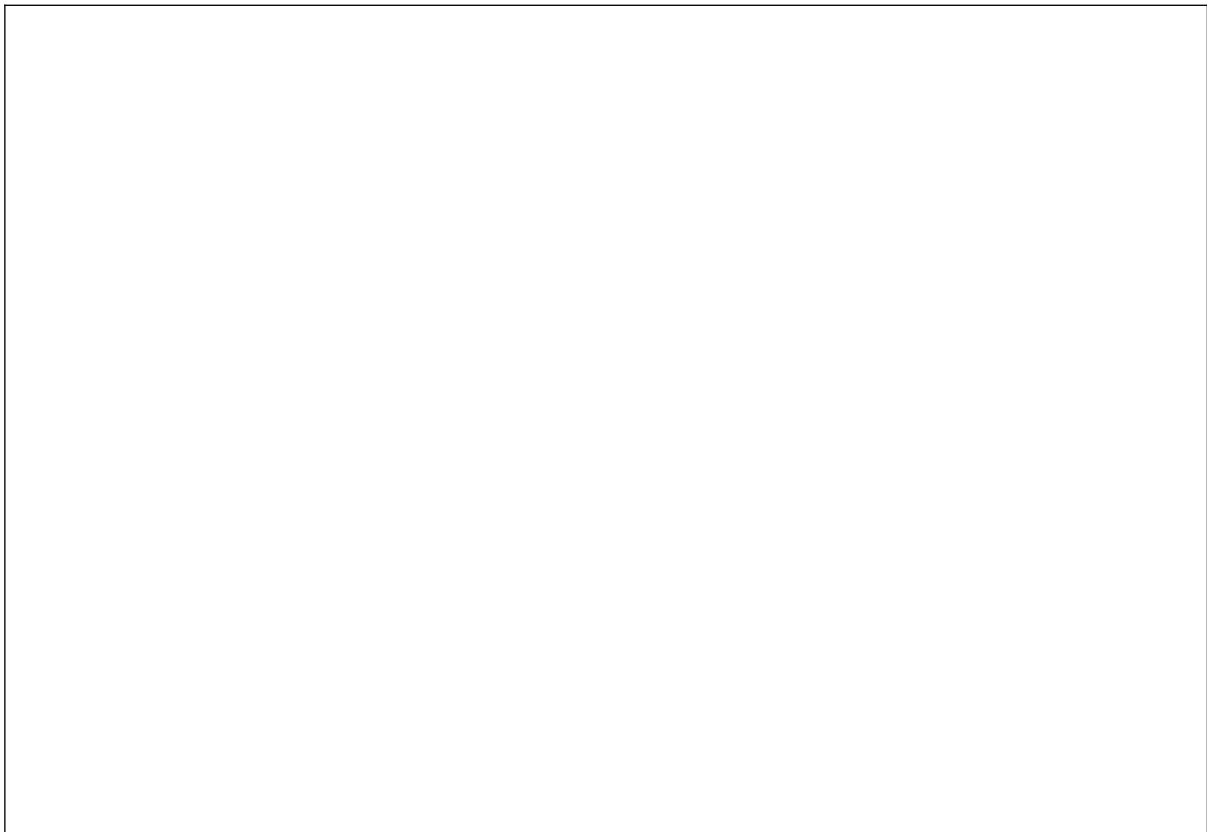
(ii) Provide RTL Schematic here.

(iii) Add your testbench here for following cases.

- a. 3+4
- b. 3-4
- c. 7+7
- d. 2-6



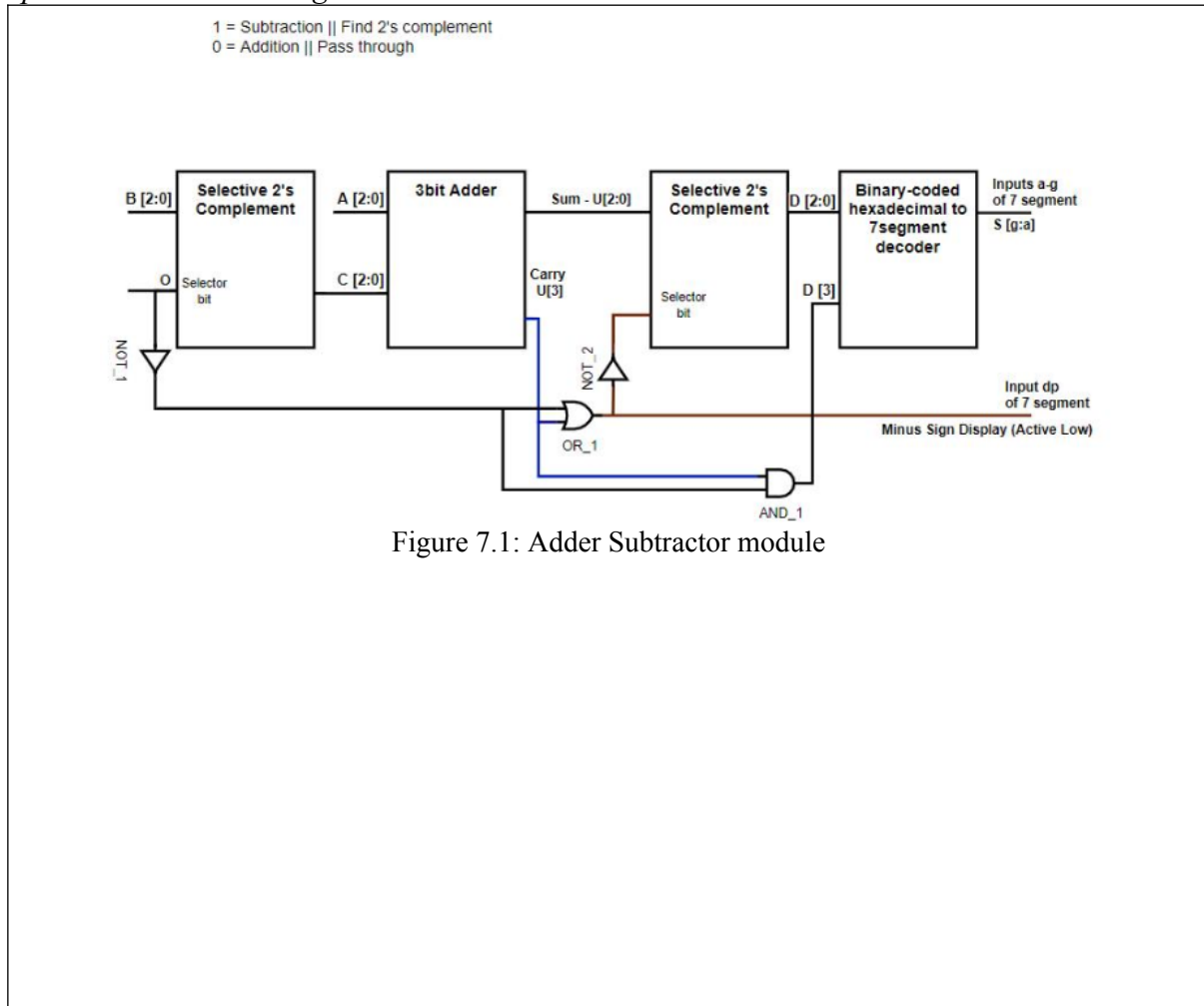
- (iv) Attach screenshot of waveform results here. Make sure to scale properly for visibility of all cases.



Exercise

Use Error: Reference source not found and work out for cases when B is zero and subtracted from A. Reflect on the output obtained and suggest ways to fix (if required).

Space to work out on Figure 7.1.



Space for reflection

Assessment Rubric Lab 07 - Worksheet Design Adder Subtractor

Name:	ID:	Section:
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Marks Distribution:

		LR2 Code	LR4 Data Collection	LR5 Results	LR10 Analysis
In - Lab	Task a	/20	/5	/10	-
	Task b	/20	-	/20	-
Exercise		-	-	-	/15
Total Marks: 100		/40	/5	/30	/15
CLO Mapping		CLO2	CLO2	CLO2	CLO2

Affective Domain Rubric		Points	CLO Mapped
AR 7	Report Submission and Viva	/10	CLO 4

CLO	Total Points	Points Obtained
2	90	
4	10	
Total	100	

For description of different levels of the mapped rubrics, please refer to the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.

Lab Evaluation Assessment Rubrics

#	Assessment Elements	Level 1: Unsatisfactory	Level 2: Developing	Level 3: Good	Level 4: Exemplary
LR2	Program/Code/ Simulation Model/ Network Model	Program/code/simulation model/network model does not implement the required functionality and has several errors. The student is not able to utilize even the basic tools of the software.	Program/code/simulation model/network model has some errors and does not produce completely accurate results. Student has limited command on the basic tools of the software.	Program/code/simulation model/network model gives correct output but not efficiently implemented or implemented by computationally complex routine.	Program/code/simulation /network model is efficiently implemented and gives correct output. Student has full command on the basic tools of the software.
LR4	Data Collection	Measurements are incomplete, inaccurate and imprecise. Observations are incomplete or not included. Symbols, units and significant figures are not included.	Measurements are somewhat inaccurate and imprecise. Observations are incomplete or vague. Major errors are there in using symbols, units and significant digits.	Measurements are mostly accurate. Observations are generally complete. Minor errors are present in using symbols, units and significant digits.	Measurements are both accurate and precise. Data collection is systematic. Observations are very thorough and include appropriate symbols, units and significant digits and task completed in due time.
LR5	Results & Plots	Figures/ graphs / tables are not developed or are poorly constructed with erroneous results. Titles, captions, units are not mentioned. Data is presented in an obscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captions and proper units are mentioned. Data presentation is systematic.
LR7	Viva	Response shows a complete lack of understanding of the assigned / completed task	Response shows shallow understanding of the assigned task	Response shows substantial understanding of the assigned task	Response shows complete understanding of the completed task. The student is able to explain all the related concepts.
LR10	Analysis	Results are not interpreted or are analyzed incorrectly.	Analysis of the obtained results is incomplete. Conclusions are not drawn.	Results are analyzed and concluded but are not well explained and justification is not provided with the help of reasoning.	Results are well analyzed supported with reasons. Good comparison is made between the theoretical and experimental results with correct and useful conclusion.